

Empirical Proof Record: VALIDATED

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Proof ID: b663061362b468852daab2929e7354aa...

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1. Claim

Across 100 randomized op sequences (20 insert ops each), the two Yjs Python backends (pycrdt, y-py) converge to identical final text in $\geq 99.0\%$ of cases. Both wrappers bind to the same Yrs Rust core; disagreement is a real bug. (CLAUDE.md §Layer 4 CRDT reconciliation: Kimera's distributed-substrate fusion relies on this law holding for the underlying CRDTs.)

- **Operationalisation:** For each of N sequences (deterministic seed), generate K insert ops with random (position, char) pairs. Apply each sequence to both backends; record (a) whether final text agrees, (b) per-backend wall-time. Aggregate the agreement rate; Wilson 95% CI on the binomial proportion.
- **Threshold:** `cross_backend_convergence_rate` ≥ 0.99 proportion
- **H0:** `cross_backend_convergence_rate` < 0.99 (at least one backend disagrees on final state in $> 1.0\%$ of cases)
- **H1:** `cross_backend_convergence_rate` ≥ 0.99 (both Yjs Python wrappers — pycrdt + y_py — converge to identical final text on the vast majority of randomized op sequences, as the shared Yrs Rust core requires)

2. Verdict

VALIDATED — observed 1.0

100/100 sequences converged across both backends (1.0000); Wilson 95% CI [0.9630, 1.0000]



Figure 1: cross_backend_convergence_rate – observed value with Wilson 95% CI.

3. Evidence

Pillar	Statistic	Value	95% CI	Libr
cross_backend_yjs_convergence	cross_backend_convergence_rate	1.0	(0.9630, 1.0000)	stats

4. Pre-registration

- Registered at: 2026-05-23T23:09:39.256641+00:00
- Config hash: eef18a4bba44d947a0033d81f07e75bf...
- Data hash: eef18a4bba44d947a0033d81f07e75bf...

5. Signature

223c8d8e820db3078448aaff4ee7519a4e9200cdcc4cb84b09f87c019c4ee025